

HW #5 - Writing Equations of Parabolas

Date _____ Period _____

Use the information provided to write the vertex form equation of each parabola.

1) $y = -x^2 - 2x + 5$

2) $y = -2x^2 + 16x - 38$

3) $y = 3x^2 - 48x + 198$

4) $y = -x^2 - 6x - 6$

Use the information provided to write the intercept form equation of each parabola. Use your graphing calculator to find the x-intercepts.

5) $y = -x^2 + 11x - 18$

6) $y = 2x^2 - 22x + 36$

7) $y = 2x^2 - 36x + 160$

8) $y = -x^2 - 9x - 8$

Use the information provided to write the vertex form equation of each parabola.

9) Vertex: $(-10, 0)$, Passes through: $(-13, -9)$

10) Vertex: $(-3, 8)$, Passes through: $(-1, -60)$

11) Vertex: $(5, 9)$, Passes through: $(7, 5)$

12) Vertex: $(-4, 5)$, Passes through: $(-6, 73)$

Use the information provided to write the intercept form equation of each parabola.

13) Opens up or down, and passes through $(5, 0)$, $(0, -60)$, and $(3, 0)$

14) Opens up or down, and passes through $(11, 7)$, $(4, 0)$, and $(10, 0)$

15) Opens up or down, and passes through $(0, 0)$, $(-1, -28)$, and $(-8, 0)$

16) Opens up or down, and passes through $(2, 8)$, $(6, 0)$, and $(8, 8)$

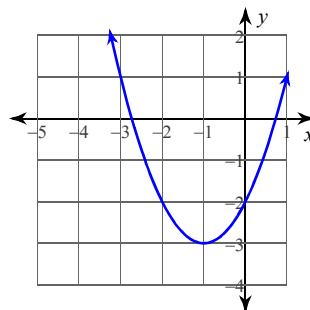
Graph each function. Find the vertex, axis of symmetry, domain, range, interval of increase, interval of decrease, and y-intercept.

17) $y = x^2 + 2x - 2$

18) $y = (x - 1)^2 + 1$

Answers to HW #5 - Writing Equations of Parabolas

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|---------------------------|----------------------------|---------------------------|--------------------------|
| 1) $y = -(x + 1)^2 + 6$ | 2) $y = -2(x - 4)^2 - 6$ | 3) $y = 3(x - 8)^2 + 6$ | 4) $y = -(x + 3)^2 + 3$ |
| 5) $y = -(x - 9)(x - 2)$ | 6) $y = 2(x - 9)(x - 2)$ | 7) $y = 2(x - 10)(x - 8)$ | 8) $y = -(x + 8)(x + 1)$ |
| 9) $y = -(x + 10)^2$ | 10) $y = -17(x + 3)^2 + 8$ | 11) $y = -(x - 5)^2 + 9$ | |
| 12) $y = 17(x + 4)^2 + 5$ | 13) $y = -4(x - 5)(x - 3)$ | 14) $y = (x - 10)(x - 4)$ | |
| 15) $y = 4x(x + 8)$ | 16) $y = (x - 6)(x - 4)$ | 17) | |



18)

